Anvil-E1 Phase 2: Official Checkpoint-Slice Integration and Software-to-RTL Evaluation

Technical Design and Verification Report

15 October 2026 (UTC+8)

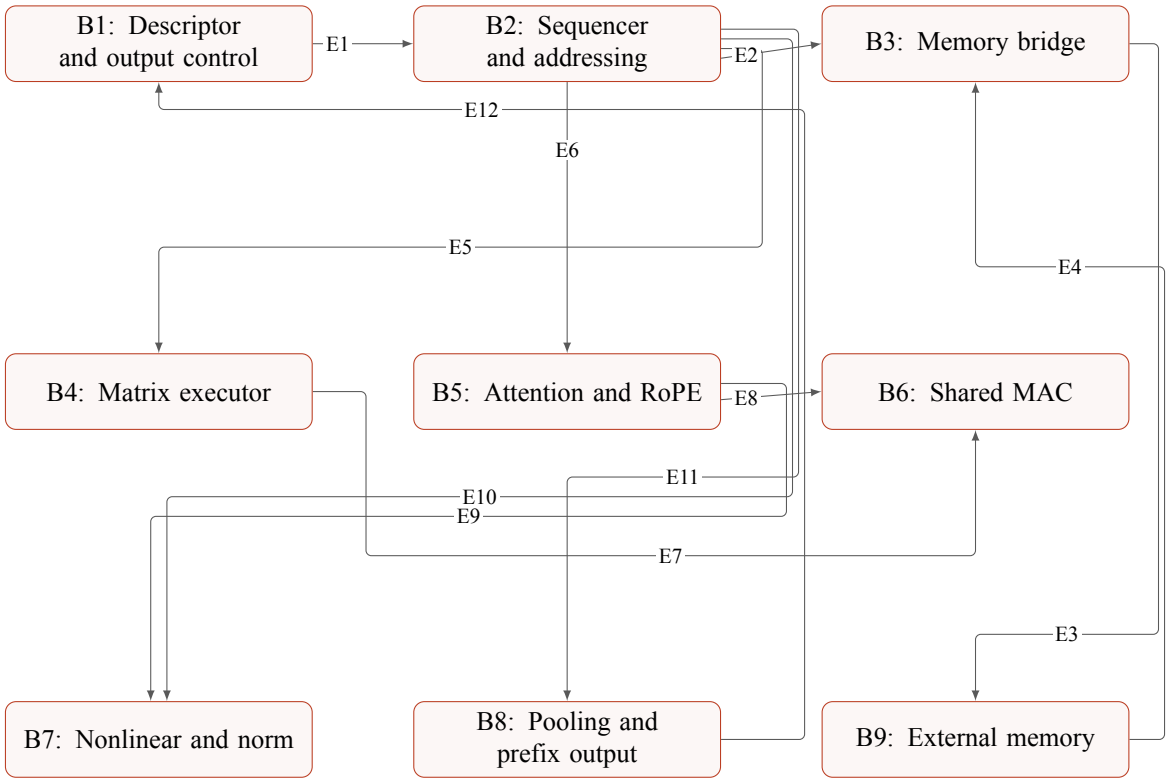
Abstract

Anvil-E1 is a modular, externally memory-backed SystemVerilog text-embedding engine implementing hybrid attention, per-layer embedding inputs, feedforward processing, masked mean pooling and Matryoshka prefix normalization. Phase 2 integrates authentic supplied Google EmbeddingGemma-2 BF16 checkpoint slices through deterministic signed-Q8.24 weight mapping and a zero-stage anvil_e_top wrapper.

All sixteen official-weight evaluations passed: 52 output words matched the independent integer oracle exactly, minimum FP32-to-RTL cosine similarity was 0.9999999998517652, and maximum absolute error was 1.2040138244628906e-5. Nine inherited configuration/seed runs, corruption sensitivity and reduced-configuration synthesis also passed. These measurements establish reduced mapped-slice execution, not full-checkpoint inference, upstream executable parity or physical performance.

1 Architecture and design rationale

Thirteen RTL files comprise twelve dedicated anvil_e1 modules and the anvil_e_top integration wrapper. The wrapper contains exactly one parameter- and port-forwarding anvil_e1_top instance named core, adding no computation, storage or latency. A hardwired sequencer shares arithmetic and memory resources across transformer operations. Weights and sequence-sized intermediates remain external; on-chip storage is bounded by vector dimensions. Attention recomputes scores across three passes instead of retaining a sequence-squared score matrix, exchanging memory traffic and latency for bounded local storage. The forward equations are adopted parent-contract conventions, not independently demonstrated upstream executable behavior. [E001, E002, E015, E016]



Functional block map, not a gate-level netlist. Edge identifiers match the directed relationships below. Crossings do not denote junctions; separate attachment points preserve arrow directions.

Block	Responsibility
B1: Descriptor and output control	Latches descriptors, validates input arrays, controls job ownership, and registers embedding and completion outputs.
B2: Sequencer and addressing	Schedules the forward path, calculates tensor and scratch addresses, and arbitrates shared-resource ownership.
B3: Memory bridge	Registers requests and responses, permitting one outstanding external transaction, including acknowledged writes.
B4: Matrix executor	Buffers an input vector, streams row-major weights and captures complete shared-MAC reductions.
B5: Attention and RoPE	Executes grouped-query hybrid attention and split-half absolute-position rotations using shared arithmetic.
B6: Shared MAC	Forms signed 64-bit products and complete 96-bit accumulations with operation-boundary rounding.
B7: Nonlinear and norm	Provides iterative nonlinear functions, weighted RMS normalization and epsilon-free L2 normalization.
B8: Pooling and prefix output	Accumulates valid-token projections, computes their mean and normalizes the requested leading channels.
B9: External memory	Integrator-owned storage for packed weights, IDs, positions, masks and sequence-sized scratch.

Directed relationship	Meaning
E1: B1 to B2	Validated job and execution ownership
E2: B2 to B3	Arbitrated memory commands
E3: B3 to B9	Registered read/write requests
E4: B9 to B3	Ordered data/error responses
E5: B2 to B4	Projection scheduling
E6: B2 to B5	Head, layer and position scheduling

Directed relationship	Meaning
E7: B4 to B6	Exclusive MAC operands and enables
E8: B5 to B6	Score and rotation products
E9: B5 to B7	Exponential and trigonometric operations
E10: B2 to B7	Normalization and nonlinear scheduling
E11: B2 to B8	Projected tokens, masks and prefix length
E12: B8 to B1	Embedding words and completion

2 Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Asserted reset zeros every public output and payload and cancels active work. The external fabric must share reset or flush obsolete responses.
job_valid, job_ready; job_tokens, job_ output_dim	Input valid and descriptor; output ready	1-bit handshake; 32-bit count; 16-bit dimension	Acceptance occurs on a rising edge with job_valid and job_ready asserted. Hold descriptor fields stable until that edge; the controller then latches them. Count is 1..MAX_SEQ_LEN; dimension selects E, 2E/3, E/3 or E/6. Further acceptance is disabled until completion retirement.
weights_base, tokens_ base, positions_base, mask_base, scratch_ base	Input	64 bits each	Four-byte-aligned byte addresses, latched with the descriptor. Hold them stable until descriptor acceptance; they need not remain fixed afterward. The host guarantees nonoverlapping regions, keeps initialized weights, IDs, positions and masks unchanged until completion, and reserves scratch for DUT writes. Hardware rejects exclusive-end address overflow. Memory responses and handshake inputs remain active protocol signals, not immutable job data. [E006, E015, E016]
mem_req_valid, mem_ req_ready, mem_req_ write, mem_req_addr, mem_req_wdata	Output request; input ready	1-bit controls; 64-bit address; 32-bit data	Registered single-word requests remain stable through stalls. Acceptance requires valid and ready on a rising edge. At most one request is outstanding.
mem_rsp_valid, mem_ rsp_ready, mem_rsp_ rdata, mem_rsp_error	Input response; output ready	1-bit controls/ error; 32-bit data	Service every accepted request, including writes, with one ordered response. Hold a valid response and its payload until acceptance. Response readiness follows request acceptance; there is no combinational zero-latency shortcut.
emb_valid, emb_ready, emb_data, emb_index, emb_last	Output stream; input ready	1-bit handshake/ last; signed 32-bit data; 16-bit index	Signed Q8.24 words use ascending zero-based indices. Data, index and last remain stable while stalled. A one-channel prefix emits index zero with last asserted. The recipient may vary emb_ready according to its protocol needs.

Signal / group	Direction	Width	Contract
done_valid, done_ ready, done_error, busy	Output status; input completion ready	1-bit handshake/ busy; 4-bit error	Completion is separately registered and held until accepted. Errors are 0 success, 1 invalid descriptor, 2 invalid token/position/mask, 3 memory error, 4 all masked and 5 address overflow. Errors emit no successful embedding. Busy remains asserted and job_ ready low through output/completion stalls; completion acceptance permits return to idle.

Parameter	Default	Meaning / supported values
VOCAB_SIZE	262144	Positive; all token IDs must be below this value, including masked entries.
HIDDEN_SIZE, PER_ LAYER_INPUT_SIZE, INTERMEDIATE_SIZE	512, 512, 2048	Each 1..65535. Per-layer input width is independently configurable.
EMBEDDING_DIM	768	6..65535 and divisible by six; default output prefixes are 768, 512, 256 and 128.
NUM_LAYERS	24	1..65535. Indices congruent to five modulo six use full attention: 5, 11, 17 and 23 at defaults.
NUM_Q_HEADS, LOCAL_KV_HEADS, FULL_KV_HEADS	4, 2, 1	Positive; query-head count is at most 65535 and divisible by each KV-head count.
LOCAL_HEAD_DIM, FULL_HEAD_DIM	256, 512	Positive even widths; each query-head-count times head-width product must not exceed 65535.
SLIDING_WINDOW, MAX_SEQ_LEN	512, 262144	Positive. Positions must be below MAX_SEQ_LEN; local attention requires absolute position distance strictly below SLIDING_WINDOW.

3 Behavior and corner cases

3.1 Official-weight mapping

Of 55 supplied tensors, 39 text tensors provide inference coefficients. Official configuration A uses (V,H,P, I,E,L,Q,localKV,fullKV,localD,fullD,window,maxSeq)=(16,4,4,8,6,6,2,1,1,2,4,2,8). Destination layers 0..4 repeat official layer 0; layer 5 uses official layer 5. Matrices select leading rows/columns using loaded source strides, not flattened prefixes. Global PLE selects rows 0..23 and columns 0..3, assigning four distinct rows per destination layer; layer 5 therefore uses available rows 20..23, not unavailable original layer-5 coordinates. Packing produces 107 destination tensors: five globals lexically, then ascending destination layer and lexical suffix, with row-major four-byte elements. No coefficients are fabricated, rescaled, neutralized or accuracy-selected. [E002, E031]

3.2 Forward computation

Token embeddings are multiplied by sqrt(H). Global PLE projection is divided by sqrt(H), reshaped per layer and unit-offset RMS-normalized. Each layer performs input RMSNorm; q/k/v projections; per-head q/k RMSNorm; RoPE and attention; output projection, post-attention norm and residual; pre-feedforward norm, tanh-GELU(gate)*up, down projection, post-feedforward norm and residual; GELU-gated PLE multiplication, projection, post-PLE norm and residual; then layer_scalar. Attention width remains Q*D until output projection. Final RMSNorm and embedding projection precede masked mean pooling, prefix selection and selected-prefix L2 normalization. Projections are bias-free and inference dropout is zero. [E001]

3.3 Attention and masks

Attention is bidirectional with contiguous grouped-query mapping. Local/full RoPE bases are 10000/1000000; rotations pair split halves at supplied absolute positions. Local keys require abs(key_position-query_position)<window; full layers see all valid keys. Explicit masks select keys and pooling contributions. There is no automatic token insertion, PAD inference or prompt exclusion. All-masked jobs fail; a nonempty zero pooled vector succeeds with zero output. [E001]

3.4 Numerical contract and correction

External weights, activations and outputs are signed Q8.24, rounded nearest/ties-even and saturated to [-128,128-2^-24]. Products are signed 64-bit and complete reductions retain at least 96 bits. Weighted RMSNorm computes x/sqrt(mean(x²)+epsilon)*(1+w), adding EPSILON_Q48=281474977 after ties-even rounding of the complete Q48 mean-square reduction. The earlier Q8.24-first epsilon, 17/2^24, caused official FP32 error-gate failures despite exact integer agreement. Direct Q48 epsilon representation resolves that denominator-precision defect without a legacy mode. L2 remains epsilon-free; nonfinite text coefficients are rejected. [E002, E011, E063]

3.5 Nonlinear and cycle boundaries

Nonlinear series use Q40 internal arithmetic. Exp uses an 18-term recurrence; sin/cos use range reduction and iterative series; GELU uses the tanh approximation and asymptotic handling at |x|>=8. Attention skips softmax differences below -16. RMS normalization retains a high-precision denominator and an 80-step integer square root. Memory-command capture, public request acceptance, response capture and consumer retirement are distinct boundaries. Embedding and completion holding stages cannot be bypassed; the wrapper adds zero stages. [E001]

4 Verification methods and measured results

4.1 Authoritative defaults, interfaces and packing

Method. Retained architecture checks examined both tops, default elaboration, address boundaries, the complete packing contract, illegal geometry and BF16 conversion/rejection.

Result. Passed: 413 expanded tensors, full layers 5/11/17/23, 65280 finite BF16 encodings checked, 256 nonfinite encodings rejected and seven illegal-parameter cases diagnosed. The retained architecture result was not rerun during execution-budget recovery.

Evidence: E052, E003, E016, E026, E033

4.2 Authentic supplied-weight consistency

Method. Streamed file SHA-256; parsed raw BF16 safetensors; validated tensor shapes, extents, ranges and hashes; independently checked supplied FP32/Q8.24 coordinates.

Result. All 55 tensor payloads passed, including excluded vision payloads; 39 text tensors supplied execution and 11266 JSON coordinates were checked. Safetensors SHA-256 is 3a310d7733a28f533b611eb7525f5973 03980c2ab2f890487a9a1fcbf0366bfb. Mapped-image SHA-256 is 2cf8db510462a09990f2d66c34598486 6d0d6374d6997b7fa4c203a9e5e1e147.

Evidence: E031, E063

4.3 Official FP32 software versus actual RTL

Method. Icarus compiled actual RTL with -g2012; vvp executed four sequences at dimensions 6/4/2/1. Software decoded raw BF16 to FP32 using struct, documented rounding boundaries and mathematical nonlinearities, independently of the integer oracle.

Result. All 16 jobs and 52 words passed with zero integer mismatches. Minimum cosine was 0.9999999998517652 against 0.999; maximum absolute error was 1.2040138244628906e-5 against 0.002. Worst per-job MAE was 6.884336471557617e-6 and RMSE 8.600952306391626e-6. Maximum absolute software/RTL norm difference was 7.76098445509632e-8 against 0.002. Official weight conversion and integer operations recorded zero saturations.

Evidence: E063, E031, E032, E039

4.4 Measured cycle-based throughput

Method. Parsed accepted public descriptor, embedding and completion edges, including stalls. Latency equals destination-edge number minus descriptor-edge number, not an inclusive edge count.

Result. For dimensions 6/4/2/1, completion cycles were [22654,22625,22620,22613] for [2]; [48400,48383,48365,48374] for [2,3]; [70669,70665,70643,70637] for [2,4,7]; and [220274,220247,220239,220232] for [2,3,4,5,6,7,8,1]. Cycles/submitted-token ranged 22613..27534.25; tokens/cycle ranged 3.6318403443e-5..4.4222349976e-5. The comparison ledger separately records first/last embedding edges, valid-token denominators and jobs/cycle.

Evidence: E063, E031

4.5 Preserved functional and primitive behavior

Method. Executed A/B/C at seeds 17/89/203 with exact scoreboards, independent quantized-input mathematical comparisons, fault/reset traffic, protocol coverage and primitive/norm gates.

Result. All nine inherited runs passed, covering full-layer indices 5/11/17/23. GELU, exp and sin/cos satisfied absolute-error limits 2e-4, 2e-6 and 2e-5; representable nonzero reciprocal/rsqrt results satisfied relative error 2e-4. Directed small/zero denominators distinguished Q48 epsilon from the historical constant and preserved zero-vector L2 behavior.

Evidence: E063, E027, E029, E034

4.6 Scoreboard corruption sensitivity

Method. Freshly compiled and executed a successful official job, validated its original trace, changed exactly one accepted embedding word by one Q8.24 LSB in a private copy, and reapplied the identical scoreboard.

Result. The original trace passed. Changing job 0/index 0 from 11924784 to 11924785 produced the specific expected/observed exact-data mismatch. The altered trace is negative-control evidence only. The inherited functional negative control also passed.

Evidence: E064, E063, E027

4.7 Synthesizable connected implementation

Method. Yosys synthesized A and B under anvil_e_top using hierarchy checking, proc, opt, memory, check -assert, statistics and retained JSON netlists.

Result. Both passed wrapper-forwarding, required-descendant, connectivity, no-latch and no-unresolved/blackbox checks. Structural traversal counted 237 sequential/235 arithmetic cells for A and 319/247 for B.

These generic structural counts are not technology area or timing measurements.

Evidence: E065, E038

5 Integration and use

Instantiate anvil_e_top and compile all rtl/*.sv; direct anvil_e1_top compatibility remains. Delivered checks require Python 3 standard library, iverilog, vvp and Yosys; NumPy is unnecessary.

Initialize packed weights and separate little-endian 32-bit ID, position and mask arrays. Reserve scratch bytes 4*N*(2*H+L*P+(Q+2*max(localKV,fullKV))*max(localD,fullD)). Scratch stores current/next hidden states, PLE and Q/K/V; do not preload hidden states or expected outputs.

Use design/pack_model.py only with a complete authoritative default checkpoint: python3 design/pack_model.py --input model.safetensors --output weights.q24.bin --inventory weights.inventory.json. Output parent directories must exist. The supplied partial official file is deliberately rejected; tests/official.py constructs the separate reduced evaluation image.

Assert reset with the memory fabric reset or flushed. Hold job_tokens, job_output_dim and all five base addresses stable until the job_valid&&job_ready accepting edge; the controller latches the descriptor. Keep initialized weights, IDs, positions and masks unchanged until completion, and reserve scratch for DUT writes. During execution, actively service ordered memory responses and output/completion ready/valid handshakes; these signals are not frozen. Consume embedding transfers, inspect done_error and retire completion through done_valid&&done_ready. [E006, E015, E016]

The recorded functional invocation is python3 tests/verify.py --check functional --output evidence/verification/functional --timeout-s 230 --configs A,B,C --seeds 17,89,203 --repair-of evidence/verification/functional/attempt-20261015T082929570124Z/receipt.json. Complete vectors, metrics, identities and cycle measurements are in evidence/verification/functional/official_weights_comparison.json; inherited results are in the producing attempt's run-ledger.json.

6 Limitations and acceptance boundary

Official execution is a deterministic reduced slice with repeated layer-0 coefficients, not original-geometry complete-checkpoint inference. In the stated geometry order, B=(20,6,4,10,12,12,4,2,1,2,4,3,8) and C=(16,4,4,8,6,24,4,2,1,2,4,2,4) extend inherited geometry coverage, not official-weight coverage.

Official cases use positions/masks [7]/[1], [1,2]/[1,1], [1,3,4]/[1,0,1], and positions 0..7 with six valid then two masked tokens. Accuracy applies to these cases, not arbitrary language workloads or embedding-retrieval quality.

Sixteen vision tensors are excluded: embed_vision.embedding_projection.weight; vision_tower.patch_embedder.input_proj.weight and position_embedding_table; and vision_tower.encoder.layers.0 suffixes input_layernorm.weight, mlp.down_proj.linear.weight, mlp.gate_proj.linear.weight, mlp.up_proj.linear.weight, post_attention_layernorm.weight, post_feedforward_layernorm.weight, pre_feedforward_layernorm.weight, self_attn.k_norm.weight, self_attn.k_proj.linear.weight, self_attn.o_proj.linear.weight, self_attn.q_norm.weight, self_attn.q_proj.linear.weight and self_attn.v_proj.linear.weight. Audio is also excluded.

Provenance establishes local consistency with supplied manifests, not independent remote retrieval. The floating path uses BF16 source coefficients with FP32 arithmetic; it does not establish BF16-activation execution or upstream executable parity.

The 10 ns testbench period is a simulation convention only. No physical clock rate, tokens/second capability, timing closure, power, technology area or default-size synthesis is established. Portable clean-directory reproduction and formal verification were not demonstrated.

A Source inventory

Production inventory: 13 Verilog/SystemVerilog files; 2510 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_e1_address.sv	173	164
rtl/anvil_e1_attention.sv	281	278
rtl/anvil_e1_controller.sv	179	175
rtl/anvil_e1_mac.sv	34	34
rtl/anvil_e1_matvec.sv	99	99
rtl/anvil_e1_memory.sv	65	65
rtl/anvil_e1_nonlinear.sv	243	237
rtl/anvil_e1_norm.sv	178	176
rtl/anvil_e1_pool.sv	162	161
rtl/anvil_e1_rope.sv	130	130
rtl/anvil_e1_sequencer.sv	778	762
rtl/anvil_e1_top.sv	158	158
rtl/anvil_e_top.sv	71	71

B Verification record index

Recorded check	Outcome	Execution time (s)
functional	passed	225.508
sensitivity	passed	3.877
synthesis	passed	94.387

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E003	design/pack_model.py
E016	rtl/anvil_e_top.sv
E026	tests/architecture.py
E027	tests/behavior.py
E029	tests/model.py
E031	tests/official.py
E032	tests/official_fp32.py
E033	tests/packing.py
E034	tests/primitives.py
E038	tests/synthesis.py
E039	tests/verify.py
E052	.vulcan/engineering/run-0002/attempt-0003/stdout.log
E063	.vulcan/engineering/run-0003/attempt-0003/stdout.log
E064	.vulcan/engineering/run-0003/attempt-0004/stdout.log
E065	.vulcan/engineering/run-0003/attempt-0005/stdout.log

C Document provenance

Engineering was independently accepted for the recorded source and evidence identity. This is not exhaustive proof, hardware deployment or physical signoff. The companion REPORT.md records the exact report approval and source manifest, and links the English/Chinese journeys containing original failures and interventions. A report number identifies a generation attempt, not a new hardware revision.

No developer attribution was supplied; none has been inferred.